

Exercise 3.6

$k, \bar{k}, \epsilon^{(1)}, \epsilon^{(2)}$ form a basis and we can also express

$P_{\mu\nu}$ in terms of $g_{\mu\nu}, k_\mu k_\nu, \bar{k}_\mu \bar{k}_\nu, \bar{k}_\mu k_\nu, k_\mu \bar{k}_\nu$.

$$P_{\mu\nu} = A g_{\mu\nu} + B \frac{k_\mu k_\nu}{k \cdot \bar{k}} + C \frac{\bar{k}_\mu \bar{k}_\nu}{k \cdot \bar{k}} + D \frac{\bar{k}_\mu k_\nu}{k \cdot \bar{k}} + E \frac{k_\mu \bar{k}_\nu}{k \cdot \bar{k}}$$

Here we have normalized such that A, B, C, D, E are dimensionless.

Because $k, \bar{k}, \epsilon^{(1)}, \epsilon^{(2)}$ form a basis they are orthogonal i.e. $\epsilon_\mu^{(i)} k^\mu = 0$ also $k^2 = \bar{k}^2 = 0$.

Let's find out the relationships between the constants A, B, C, D using projection of

$P_{\mu\nu}$ along the basis vectors:

We start with projection along $g_{\mu\nu}$:

$$k_\mu g^{\mu\nu} = k^\nu$$

$$g_{\mu\nu} P^{\mu\nu} = 4A + D \underbrace{g_{\mu\nu} \frac{\bar{k}_\mu k_\nu}{k \cdot \bar{k}}}_{=1} + E \underbrace{g_{\mu\nu} \frac{k_\mu \bar{k}_\nu}{k \cdot \bar{k}}}_{=1} = 4A + D + E = g_{\mu\nu} P^{\mu\nu}$$

$$k_\mu P^{\mu\nu} = A k_\mu g^{\mu\nu} + 0 + C \frac{k \cdot \bar{k}}{k \cdot \bar{k}} \bar{k}^\nu + D k^\nu + 0 = A k^\nu + C \bar{k}^\nu + D k^\nu$$

$$\bar{k}_\mu P^{\mu\nu} = A \bar{k}_\mu g^{\mu\nu} + B k^\nu + 0 + 0 + E \bar{k}^\nu = A \bar{k}^\nu + B k^\nu + E \bar{k}^\nu$$

$$k_\nu P^{\mu\nu} = A k_\nu g^{\mu\nu} + 0 + C \bar{k}^\mu + 0 + E k^\mu = A k^\mu + C \bar{k}^\mu + E k^\mu$$

$$\bar{k}_\nu P^{\mu\nu} = A \bar{k}_\nu g^{\mu\nu} + B k^\mu + 0 + D \bar{k}^\mu + 0 = A \bar{k}^\mu + B k^\mu + D \bar{k}^\mu$$

$$k_\mu \bar{k}_\mu P^{\mu\nu} = A k_\mu \bar{k}_\mu g^{\mu\nu} + E k \cdot \bar{k} = A k \cdot \bar{k} + E k \cdot \bar{k}$$

$$k_\mu \bar{k}_\nu P^{\mu\nu} = A k_\mu \bar{k}_\nu g^{\mu\nu} + D k_\mu \bar{k}^\mu = (A + D) k \cdot \bar{k}$$

Should be same under change of indices
 $\Rightarrow E = D$

$$k_\mu k_\nu P^{\mu\nu} = C k \cdot \bar{k} \quad \text{and} \quad \bar{k}_\mu \bar{k}_\nu P^{\mu\nu} = B k \cdot \bar{k} = (k^\rho k^\sigma g_{\rho\mu} g_{\sigma\nu} P^{\mu\nu})$$

$$\Rightarrow B = C \quad = (k^\rho k^\sigma P_{\rho\sigma}) = (k_\rho k_\sigma P^{\rho\sigma}) = C \cdot k \cdot \bar{k}$$

Additionally: $k_\mu P^{\mu\nu} = k_\mu \sum_\lambda \epsilon_\lambda^\mu \epsilon_\lambda^\nu = 0$ // since $k_\mu \epsilon_\lambda^\mu = 0$

$$\Rightarrow \bar{k}_\mu P^{\mu\nu} = 0$$

Now we can relate the results of projecting $P^{\mu\nu}$ in 2-different way:

$$\text{From } g_{\mu\nu} P^{\mu\nu} = 4A + D + E = 4A + 2D \quad \text{and} \quad g_{\mu\nu} P^{\mu\nu} = g_{\mu\nu} \sum_{\lambda} \varepsilon_{\lambda}^{\mu} \varepsilon_{\lambda}^{\nu *} = \sum_{\lambda} \underbrace{\varepsilon_{\lambda} \cdot \varepsilon_{\lambda}}_{=-1} = -2$$

$$\text{we get } 4A + 2D = -2 \Rightarrow 2A + D = -1$$

$$\text{From } k_{\mu} P^{\mu\nu} = 0 = (A+D)k^{\nu} + C\bar{k}^{\nu} \quad \text{we get } A = -D \quad \text{and} \quad C = B = 0$$

$$\Rightarrow 2A + D = -1 \Leftrightarrow 2A - A = -1 \Leftrightarrow A = -1 \Rightarrow D = 1$$

Now we apply the results to the general P :

$$P_{\mu\nu} = \sum_{\lambda} \varepsilon_{\mu}^{\lambda} \varepsilon_{\nu}^{\lambda *} = A g_{\mu\nu} + B \left[\frac{k_{\mu} k_{\nu} + \bar{k}_{\mu} \bar{k}_{\nu}}{k \cdot \bar{k}} \right] + D \left[\frac{\bar{k}_{\mu} k_{\nu} + k_{\mu} \bar{k}_{\nu}}{k \cdot \bar{k}} \right]$$

$$\Rightarrow \underline{P_{\mu\nu} = \sum_{\lambda} \varepsilon_{\mu}^{\lambda} \varepsilon_{\nu}^{\lambda *} = -g_{\mu\nu} + \frac{\bar{k}_{\mu} k_{\nu} + k_{\mu} \bar{k}_{\nu}}{k \cdot \bar{k}}} \quad \square$$

Exercise 3.7

Consider pure electrodynamics,

$$S[A] = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right)$$

using the gauge fixing condition $G_w[A] = \partial_{\mu} A^{\mu} + \frac{1}{2} \eta A_{\mu} A^{\mu} - w$,

where $w(x) = w$ is an arbitrary function.

$$\text{The FP-determinant is } \Delta_{\text{FP}} = \det \left(\frac{\delta G_w[A^*]}{\delta \alpha} \right)$$

$$\Rightarrow 1 = \int D\alpha(x) \delta(G[A^*]) \det \left(\frac{\delta G_w[A^*]}{\delta \alpha} \right)$$

We insert this into the definition of path integral to get

$$\begin{aligned} \int D\alpha \int DA_{\mu} \Delta_{\text{FP}}(A_{\mu}) \delta(G[A_{\mu}^*]) e^{iS[A]} &= \int D\alpha \int DA_{\mu} e^{iS[A]} \delta(G[A^*]) \det \left(\frac{\delta G[A^*]}{\delta \alpha} \right) \\ &= \int D\alpha \int DA_{\mu} e^{iS[A]} \delta \left(\partial_{\mu} A^{\mu} + \frac{1}{2} \eta A_{\mu} A^{\mu} - w \right) \det \left(\frac{\delta G[A^*]}{\delta \alpha} \right) \end{aligned}$$

This integral is true for any $w(x)$ and $\det \left(\frac{\delta G[A^*]}{\delta \alpha} \right)$ is independent of w so

we can multiply by normalization constant and at the same time integrate over $w(x)$ with

a Gaussian weight:

$$\Rightarrow N(\xi) \int D\omega \exp\left[-i \int d^4x \frac{\omega^2}{2\xi}\right] \int D\alpha \int DA_\mu e^{iS[A]} \delta(\partial_\mu A^{\mu*} + \frac{1}{2}\eta \tilde{A}_\mu \tilde{A}^{\mu*} - \omega(x)) \det\left(\frac{\delta G[A^*]}{\delta \alpha}\right)$$

$$= N(\xi) \int D\alpha \int DA_\mu e^{iS[A]} \exp\left[-i \int d^4x \frac{1}{2\xi} \left[\partial_\mu A^{\mu*} + \frac{1}{2}\eta \tilde{A}_\mu \tilde{A}^{\mu*}\right]^2\right] \det\left[\frac{\delta G[A^*]}{\delta \alpha}\right]$$

Since the Lagrangian is invariant under gauge transformation we can replace A in $e^{iS[A]}$ by A^* . This doesn't change the measure since $DA_\mu = DA_\mu^*$ since $A \rightarrow A^*$ is a simple shift.

$$\Rightarrow = N(\xi) \int D\alpha \int DA_\mu e^{iS[A^*]} \exp\left[-i \int d^4x \frac{1}{2\xi} \left[\partial_\mu A^{\mu*} + \frac{1}{2}\eta \tilde{A}_\mu \tilde{A}^{\mu*}\right]^2\right] \det\left[\frac{\delta G[A^*]}{\delta \alpha}\right]$$

Now we use identity $\int D\bar{c} Dc \exp\left[-i \int d^4x \bar{c} \left(\frac{\delta G[A^*]}{\delta \alpha}\right) c\right] = \det\left[\frac{\delta G[A^*]}{\delta \alpha}\right]$

Which gives us:

$$N(\xi) \int D\alpha DA_\mu D\bar{c} Dc \exp\left[i \int d^4x \mathcal{L}_{QED} - \frac{1}{2\xi} (\partial_\mu A^{\mu*} + \frac{1}{2}\eta \tilde{A}_\mu \tilde{A}^{\mu*})^2 - \bar{c} \left(\frac{\delta G[A^*]}{\delta \alpha}\right) c\right]$$

Let's now calculate $\frac{\delta G[A^*]}{\delta \alpha}$ with $G[A^*] = \partial_\mu \tilde{A}^{\mu*} + \frac{1}{2}\eta \tilde{A}_\mu \tilde{A}^{\mu*} - \omega$ $\parallel A_\mu^* = A_\mu + \frac{1}{e} \partial_\mu \alpha$

$$\frac{\delta G[A^*]}{\delta \alpha} = \frac{\delta}{\delta \alpha} \left[\partial_\mu A^{\mu*} + \frac{1}{e} \partial_\mu \partial^\mu \alpha + \frac{1}{2}\eta \left[(A_\mu + \frac{1}{e} \partial_\mu \alpha) (A^\mu + \frac{1}{e} \partial^\mu \alpha) \right] \right]$$

$$= \frac{1}{e} \partial^2 + \frac{1}{2}\eta \frac{\delta}{\delta \alpha} \left[A_\mu A^\mu + \frac{1}{e} \underbrace{(\partial_\mu \alpha) A^\mu + \frac{1}{e} A_\mu (\partial^\mu \alpha)}_{= \frac{2}{e} A_\mu (\partial^\mu \alpha)} + \frac{1}{e^2} \underbrace{\partial_\mu \alpha \partial^\mu \alpha}_{= \frac{1}{2} \partial^2 \alpha^2} \right] = \frac{1}{e} \partial^2 + \frac{1}{2}\eta \frac{\delta}{\delta \alpha} \left[A_\mu A^\mu + \frac{1}{e} \partial_\mu \alpha \partial^\mu \alpha \right]$$

$$= \frac{1}{e} \partial^2 + \frac{1}{e} \eta A_\mu \partial^\mu + \frac{1}{2e^2} \eta \partial^2 \alpha$$

, we have α -dependence because the Gauge is non-linear

Our Path integral has become:

Standard QED-Lagrangian

$$\Rightarrow F \equiv N(\xi) \int D\alpha DA_\mu D\bar{c} Dc \exp\left[i \int d^4x \mathcal{L}_{QED} - \frac{1}{2\xi} (\partial_\mu A^{\mu*} + \frac{1}{2}\eta \tilde{A}_\mu \tilde{A}^{\mu*})^2 - \bar{c} \left(\frac{1}{e} \partial^2 [1 + \frac{1}{2e} \eta \alpha] + \frac{1}{e} \eta A_\mu \partial^\mu \right) c \right]$$

With the Help of term $\frac{1}{2\xi} (\partial_\mu A^{\mu*} + \frac{1}{2}\eta \tilde{A}_\mu \tilde{A}^{\mu*})^2$ we can construct a photon Propagator

and $\frac{1}{e} \partial^2 [1 + \frac{1}{2e} \eta \alpha]$ is responsible for ghost propagator and $\frac{1}{e} \eta A_\mu \partial^\mu$ is responsible for a interaction between photon field and Ghost-field. Let's now write just the Lagrangian

part and read off the propagators:

$$\mathcal{L} = \mathcal{L}_{\text{QED}} - \frac{1}{2\xi} (\partial_\mu A^\mu + \frac{1}{2e}\eta A^\mu)^2 - \bar{c} [\partial^2 (1 + \frac{1}{2e}\eta\alpha) + \eta A_\mu \partial^\mu] c$$

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}} + \frac{1}{2\xi} (\partial_\mu A^\mu + \frac{1}{2e}\eta A^\mu)^2 - \bar{c} [\partial^2 (1 + \frac{1}{2e}\eta\alpha) + \eta A_\mu \partial^\mu] c$$

here $\mathcal{L}_0 = \mathcal{L}_{\text{free}}$ and $\mathcal{L}_{\text{int}}$ is interaction Lagrangian

The photon only part is:

$$A_\mu (\partial^2 g^{\mu\nu} - \partial^\mu \partial^\nu A_\nu) - \frac{1}{2\xi} (\underbrace{\partial_\mu A^\mu}_{=A^\mu \partial^\mu} + \frac{1}{2e}\eta A^\mu) (\underbrace{\partial^\nu A_\nu}_{=A^\nu \partial^\nu})$$

$$= A_\mu (\partial^2 g^{\mu\nu} - \partial^\mu \partial^\nu A_\nu) - \frac{1}{2\xi} (A_\mu [\partial^\mu \partial^\nu + \frac{1}{2e}\eta \partial^\mu A^\nu] A_\nu + \frac{1}{2e}\eta A_\mu [A^\mu \partial^\nu + \frac{1}{2e}\eta A^\mu A^\nu] A_\nu)$$

$$= A_\mu [\partial^2 g^{\mu\nu} - \partial^\mu \partial^\nu - \frac{1}{\xi} \partial^\mu \partial^\nu - \frac{1}{2\xi} \eta \partial^\mu A^\nu - \frac{1}{2\xi} \eta A^\mu \partial^\nu - \frac{1}{4\xi} \eta^2 A^\mu A^\nu] A_\nu$$

$$= A_\mu [\partial^2 g^{\mu\nu} - (1 - \frac{1}{\xi}) \partial^\mu \partial^\nu] A_\nu - \frac{\eta}{2\xi} A_\mu [\partial^\mu A^\nu - A^\mu \partial^\nu] A_\nu - \frac{1}{4\xi} \eta^2 \underbrace{A_\mu A^\mu A^\nu A_\nu}_{=A^4}$$

$$= \underbrace{A_\mu [\partial^2 g^{\mu\nu} - (1 - \frac{1}{\xi}) \partial^\mu \partial^\nu] A_\nu}_{\text{This term lead to the propagator and this is exactly same as when the gauge-fixing was linear in } A_\mu} - \underbrace{\frac{\eta}{2\xi} A_\mu [\partial^\mu A^\nu - A^\mu \partial^\nu] A_\nu - \frac{1}{4\xi} \eta^2 A^4}_{\text{Self-interaction terms that arise from non-linear gauge-fixing}}$$

This term lead to the

propagator and this is exactly

same as when the gauge-fixing was linear in A_μ

Self-interaction terms that arise from non-linear gauge-fixing

$\Rightarrow$ The photon propagator is still the same:

$$\tilde{D}_F(k) = \frac{-i}{k^2 + i\epsilon} (g^{\mu\nu} - (1 - \xi) \frac{k^\mu k^\nu}{k^2})$$

The ghost Lagrangian: $\mathcal{L}_{\text{ghost}} = \bar{c} [-\partial^2 (1 + \frac{1}{2e}\eta\alpha) - \eta A_\mu \partial^\mu] c$

We can read from this Lagrangian that the ^{ghost} propagator is $i \frac{1 + \frac{1}{2e}\eta\alpha}{p^2}$

$$\Rightarrow \text{---} \text{---} \text{---} \text{---} = i \frac{1 + \frac{\eta}{2e}\alpha}{p^2}$$

so the ghost propagator depends on α that is a gauge-transformation variable.

The full QED Lagrangian in this gauge-fixing is

$$\mathcal{L} = \bar{\Psi} (i\not{D} - m) \Psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial_\mu A^\mu + \frac{1}{2e}\eta A^\mu)^2 - \bar{c} [\partial^2 (1 + \frac{\eta\alpha}{2e}) + \eta A_\mu \partial^\mu] c$$

Here we have absorbed one of the $\frac{1}{e}$ -term into $\bar{c}$ and c definition.

Let's absorb ξ into definition of $\xi \Rightarrow 2\xi \rightarrow \xi$

Exercise 3.8

$$\mathcal{L} = \mathcal{L}_{QED} - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 + \partial^\mu \bar{c} \partial_\mu c$$

where $\mathcal{L}_{QED}$ is $\mathcal{L}_{QED} = \bar{\Psi}(i\not{D} - m)\Psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}$, with $D_\mu = (\partial_\mu + ieA_\mu)$

$$\Rightarrow \mathcal{L} = \bar{\Psi}(i\not{D} - m)\Psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e\bar{\Psi}\gamma^\mu \Psi A_\mu - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 + \partial^\mu \bar{c} \partial_\mu c$$

Let's start doing the transformation part by part: first we only transform A_μ then c and $\bar{c}$ then Ψ and $\bar{\Psi}$ and after this we have done the whole transformation:

The transformation we are aiming to do is ($\delta\eta$ is Grassmann parameter)

$$\left. \begin{aligned} A_\mu(x) &\rightarrow A'_\mu(x) = A_\mu(x) + \partial_\mu c(x) \delta\eta \\ c(x) &\rightarrow c'(x) = c(x) \\ \bar{c}(x) &\rightarrow \bar{c}'(x) = \bar{c}(x) + \frac{1}{\xi} \partial^\mu A_\mu(x) \delta\eta \end{aligned} \right\} \begin{aligned} \Psi(x) &\rightarrow \Psi'(x) = \Psi(x) - ie c(x) \delta\eta \Psi(x) \\ \bar{\Psi}(x) &\rightarrow \bar{\Psi}'(x) = \bar{\Psi}(x) + ie \bar{\Psi}(x) c(x) \delta\eta \end{aligned}$$

Under this transformation the Lagrangian becomes:

$\Rightarrow$

$$\mathcal{L} \rightarrow \mathcal{L}' = \bar{\Psi}'(i\not{D} - m)\Psi' - \frac{1}{4} F'_{\mu\nu} F'^{\mu\nu} - e\bar{\Psi}'\gamma^\mu \Psi' A'_\mu - \frac{1}{2\xi} (\partial_\mu A'^\mu)^2 + \partial^\mu \bar{c} \partial_\mu c$$

First let's only expand A'_μ in terms of A_μ

$$\Rightarrow F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu = F_{\mu\nu} + \partial_\mu \partial_\nu c \delta\eta - \partial_\nu \partial_\mu c \delta\eta$$

$$\Rightarrow F'_{\mu\nu} F'^{\mu\nu} = F_{\mu\nu} F^{\mu\nu} + F_{\mu\nu} \partial^\mu \partial^\nu c \delta\eta - F_{\mu\nu} \partial^\nu \partial^\mu c \delta\eta + \partial_\mu \partial_\nu c \delta\eta F^{\mu\nu} - \partial_\nu \partial_\mu c \delta\eta F^{\mu\nu}$$

All the other terms are 0 since they are proportional to $(\delta\eta)^2 = 0$

We can change the order of derivation freely because for a smooth-enough function

$\partial_\nu \partial_\mu f(x) = \partial_\mu \partial_\nu f(x)$, and we assume all our function are smooth

$$\Rightarrow F_{\mu\nu} [\partial^\mu \partial^\nu c \delta\eta - \partial^\nu \partial^\mu c \delta\eta] = 0 \quad \text{and} \quad [\partial_\mu \partial_\nu - \partial_\nu \partial_\mu] c \delta\eta F^{\mu\nu} = 0$$

$$\Rightarrow F'_{\mu\nu} F'^{\mu\nu} = F_{\mu\nu} F^{\mu\nu} \quad \Rightarrow F_{\mu\nu} F^{\mu\nu} \text{ is invariant under the transformation}$$

For term $e\bar{\Psi}'\gamma^\mu \Psi' A'_\mu$ we get $e\bar{\Psi}'\gamma^\mu \Psi' A'_\mu = e\bar{\Psi}'\gamma^\mu \Psi' A_\mu + e\bar{\Psi}'\gamma^\mu \Psi' \partial_\mu c \delta\eta$

and for $(\partial_\mu A'^\mu)^2$ we get $(\partial_\mu A'^\mu)^2 = (\partial_\mu A^\mu + \partial_\mu \partial^\mu c \delta\eta)^2 = (\partial_\mu A^\mu)^2 + 2\partial_\mu A^\mu \partial^\mu c \delta\eta + \partial^\mu \partial^\mu c \delta\eta \partial_\mu A^\mu$

$$\Rightarrow (\partial_\mu A'^\mu)^2 = (\partial_\mu A^\mu)^2 + 2\partial_\mu A^\mu \partial^\mu c \delta\eta + \partial^\mu \partial^\mu c \delta\eta \partial_\mu A^\mu$$

So our Lagrangian is now: $\mathcal{L}' = \bar{\Psi}'(i\not{D} - m)\Psi' - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e\bar{\Psi}'\gamma^\mu \Psi' A_\mu + e\bar{\Psi}'\gamma^\mu \Psi' \partial_\mu c \delta\eta - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 - \frac{1}{2\xi} [(\partial_\mu A^\mu) \partial^\mu c \delta\eta + (\partial^\mu \partial^\mu c \delta\eta) \partial_\mu A^\mu] + \partial^\mu \bar{c} \partial_\mu c$

Now let's expand $\bar{c}'$ and c'

let's change the indices to not get confused

$$\rightarrow \partial^\mu \bar{c}' \partial_\mu c' = \partial^\mu \left(\bar{c} + \frac{1}{\xi} \partial^\nu A_\nu(x) \delta\eta \right) \partial_\mu c = \partial^\mu \bar{c} \partial_\mu c + \frac{1}{\xi} \partial^\mu (\partial^\nu A_\nu \delta\eta) \partial_\mu c$$

$$= \partial^\mu [\partial^\nu A_\nu \delta\eta \partial_\mu c] - (\partial^\nu A_\nu) \delta\eta \partial^2 c$$

$$\Rightarrow \mathcal{L}' = \bar{\Psi}' (i\not{\partial} - m) \Psi' - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e \bar{\Psi}' \gamma^\mu \Psi' A_\mu - e \bar{\Psi}' \gamma^\mu \Psi' \partial_\mu c \delta\eta - \frac{1}{2\xi} (\partial_\mu A^\mu)^2$$

$$- \frac{1}{2\xi} \left[(\partial_\mu A^\mu) (\partial^2 c \delta\eta) + (\partial^2 c \delta\eta) (\partial_\mu A^\mu) \right] + \partial^\mu \bar{c} \partial_\mu c + \frac{1}{\xi} \left(\partial^\mu [\partial^\nu A_\nu \delta\eta \partial_\mu c] - (\partial^\nu A_\nu) \delta\eta \partial^2 c \right)$$

The term $\partial^\mu [\partial^\nu A_\nu \delta\eta \partial_\mu c]$ is a boundary term so it vanishes in the path integral in the exponent under the integral $\int d^4x \Rightarrow$ no actual contribution to the Lagrangian $\Rightarrow$ we can set it to zero.

$$\Rightarrow \mathcal{L}' = \bar{\Psi}' (i\not{\partial} - m) \Psi' - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e \bar{\Psi}' \gamma^\mu \Psi' A_\mu - e \bar{\Psi}' \gamma^\mu \Psi' \partial_\mu c \delta\eta - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 + \partial^\mu \bar{c} \partial_\mu c$$

$$- \frac{1}{\xi} (\partial_\nu A^\nu) \partial^2 c \delta\eta - \frac{1}{\xi} (\partial_\nu A^\nu) \delta\eta \partial^2 c$$

$$\left\| \begin{array}{l} -\frac{1}{\xi} (\partial_\nu A^\nu) [\partial^2 c \delta\eta + \delta\eta \partial^2 c] = 0 \\ \text{since } \delta\eta \text{ and } c \text{ are Grassmannian} \end{array} \right.$$

Now we expand $\bar{\Psi}'$ and Ψ'

$$\Rightarrow \bar{\Psi}' (i\not{\partial} - m) \Psi' = [\bar{\Psi}(x) + ie\bar{\Psi}(x)c(x)\delta\eta] (i\not{\partial} - m) [\Psi - ie c \delta\eta \Psi]$$

$$= \bar{\Psi} (i\not{\partial} - m) \Psi - ie\bar{\Psi} (i\not{\partial} - m) c \delta\eta \Psi + ie\bar{\Psi} c \delta\eta (i\not{\partial} - m) \Psi$$

$$= \bar{\Psi} (i\not{\partial} - m) \Psi - ie\bar{\Psi} [i\gamma^\mu (\partial_\mu c) \delta\eta \Psi + i\gamma^\mu c \delta\eta (\partial_\mu \Psi) - m c \delta\eta \Psi] + ie\bar{\Psi} c \delta\eta (i\not{\partial} - m) \Psi$$

$$\left\| \begin{array}{l} c, \delta\eta, \Psi \text{ are Grassmann variables} \\ \Rightarrow c \delta\eta \Psi = -c \Psi \delta\eta = \Psi c \delta\eta \end{array} \right.$$

$$= \bar{\Psi} (i\not{\partial} - m) \Psi + e\bar{\Psi} \gamma^\mu \Psi \partial_\mu c \delta\eta - ie\bar{\Psi} c \delta\eta (i\not{\partial} - m) \Psi + ie\bar{\Psi} c \delta\eta (i\not{\partial} - m) \Psi$$

$$= \bar{\Psi} (i\not{\partial} - m) \Psi + e\bar{\Psi} \gamma^\mu \Psi \partial_\mu c \delta\eta$$

$$\Rightarrow \bar{\Psi}' \gamma^\mu \Psi' = (\bar{\Psi} + ie\bar{\Psi} c \delta\eta) \gamma^\mu (\Psi - ie c \delta\eta \Psi) = \bar{\Psi} \gamma^\mu \Psi + \underbrace{ie\bar{\Psi} c \delta\eta \gamma^\mu \Psi - ie\bar{\Psi} \gamma^\mu c \delta\eta \Psi}_{=0}$$

$$\Rightarrow \bar{\Psi}' \gamma^\mu \Psi' = \bar{\Psi} \gamma^\mu \Psi \Rightarrow \bar{\Psi} \gamma^\mu \Psi \text{ is invariant}$$

$$\Rightarrow \mathcal{L}' = \bar{\Psi} (i\not{\partial} - m) \Psi + e\bar{\Psi} \gamma^\mu \Psi \partial_\mu c \delta\eta - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - e \bar{\Psi} \gamma^\mu \Psi A_\mu - e \bar{\Psi} \gamma^\mu \Psi \partial_\mu c \delta\eta + \frac{1}{2\xi} (\partial_\mu A^\mu)^2 + \partial^\mu \bar{c} \partial_\mu c$$

$$\mathcal{L}' = \bar{\Psi} (i\not{\partial} - m) \Psi - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 + \partial^\mu \bar{c} \partial_\mu c = \mathcal{L}_{QED} - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 + \partial^\mu \bar{c} \partial_\mu c$$

$$\Rightarrow \mathcal{L}' = \mathcal{L} \Rightarrow \text{The Lagrangian } \mathcal{L} \text{ is invariant under the BRS transformations.}$$